

Artificial Intelligence

Course Code: CS-352

Topic: Introduction to Problem Solving Agents in AI

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Week-2 CLO

This week lecture will cover the following CLO:

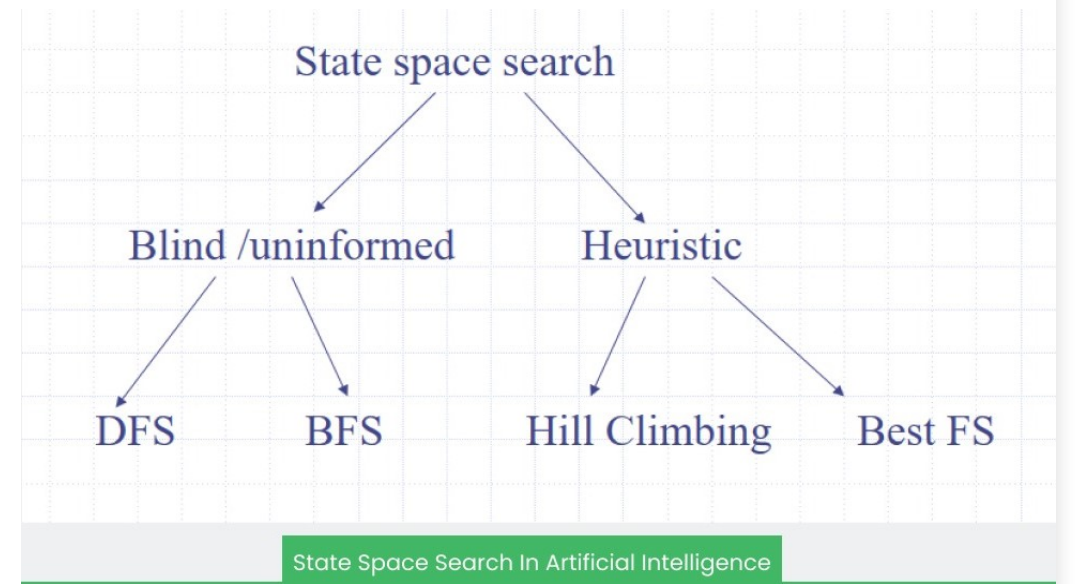
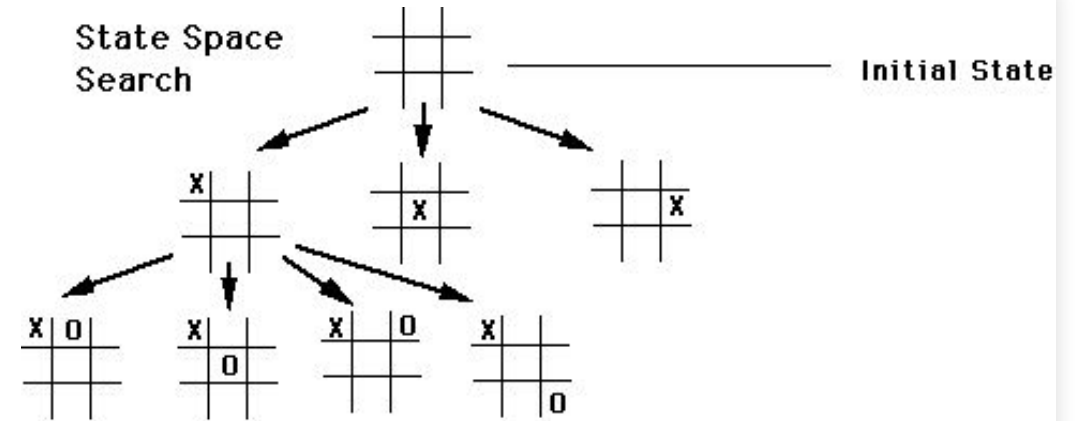
- **CLO1:** Discuss the key concepts in the field of artificial intelligence

Introduction to Problem-Solving Agents in AI

- **Definition:** An agent that takes actions to achieve a goal by finding a solution to a given problem.
- Problem-solving involves searching through a sequence of possible actions.
- The agent operates within a state space, where each state represents a configuration of the world.

State- Space

- In AI, **state space** refers to the **set of all possible states** that a **system can be in**.
- It is used in search problems, game playing, planning, and other AI applications.
- State space helps AI models determine the best path from the initial state to the goal state using search algorithms



State- Space

State: A specific configuration of the system at a given time.

Initial State: The starting point of the problem.

Goal State: The desired outcome or solution.

Operators (Actions): The possible moves or transitions that change one state to another.

State Space Graph: A representation of all states and their connections through valid transitions.



Find state space

- Route Planning (Navigation / Shortest Path)
- Vacuum Cleaner World (two-room example)

Route Planning (Navigation / Shortest Path)

- **State:** The current node (city / intersection) where the agent is located.
- **Initial State:** The start city/node (e.g., City A).
- **Goal State:** The destination city/node (e.g., City Z)
- **Operators:** Travel along an edge (road) from the current node to an adjacent node.
- **State Space:** All nodes in the road/graph.

Vacuum Cleaner World (two-room example)

- **State:** (Location, Dirty_A?, Dirty_B?) — for two rooms A and B, $\text{Location} \in \{A, B\}$, and each room has a boolean dirty/clean flag. Example: (A, Dirty, Clean).
- **Initial State:** Example (A, Dirty, Dirty) (agent in A and both rooms dirty), or any specified starting combination.
- **Goal State:** Both rooms clean (i.e., Dirty_A? = False and Dirty_B? = False), location can be either A or B unless goal requires a specific place.
- **Operators:**
 - Suck — if current location is dirty, make it clean.
 - MoveToOtherRoom (move left/right) — change Location to the other room.
- **State Space:** For two rooms there are 2 possible locations $\times 2^2 = 4$ dirt combinations = 8 possible states total.

Problem-Solving Methods

What is meant by Problem-Solving method?

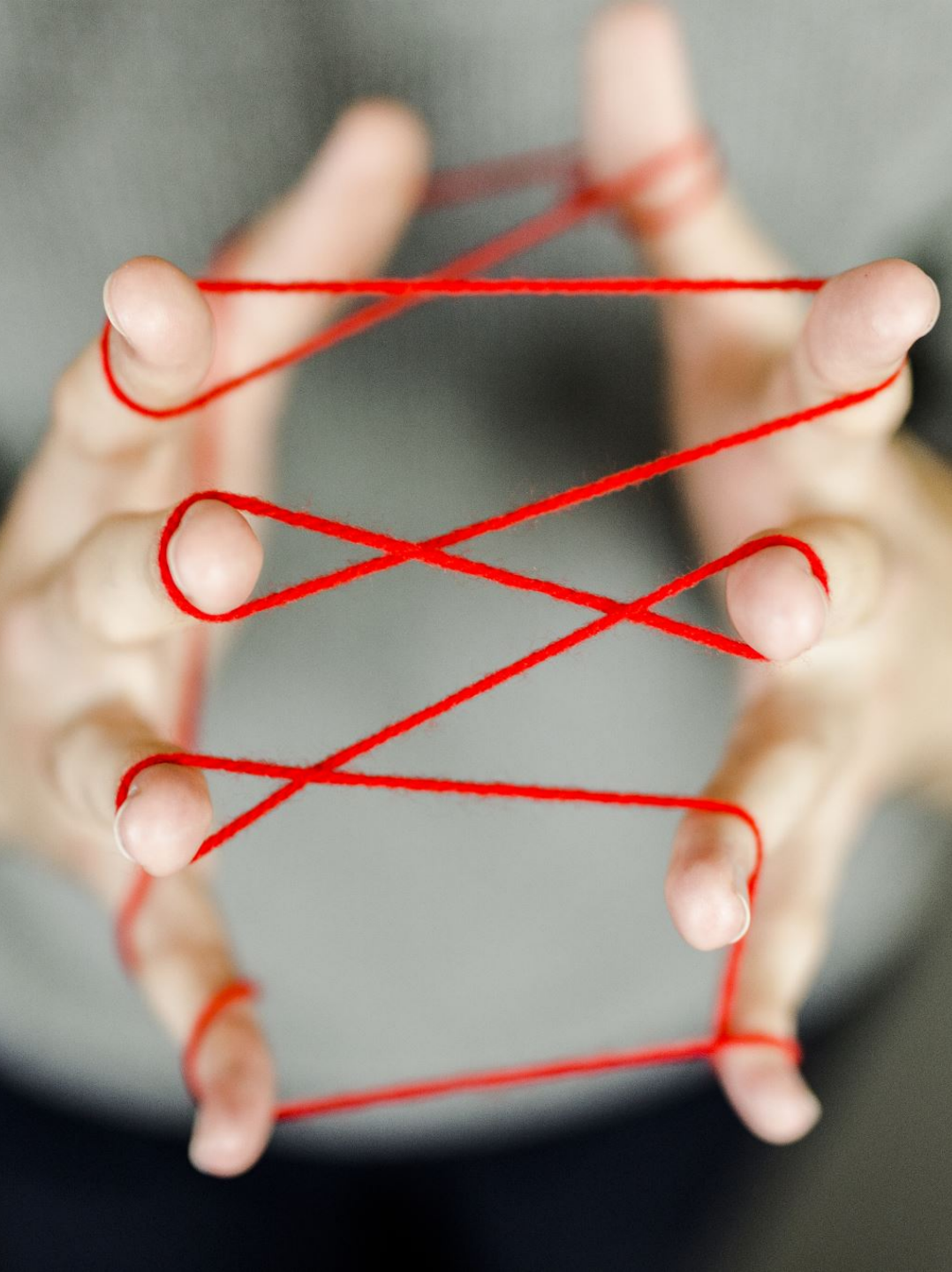
- The method of solving problem through AI involves the process of defining the search space, deciding start and goal states then finding the path from state to goal state through search space.
- The movement from start state to goal state is guided by set of rules specifically designed for that problem.

Problem

- It is the question which is to solved. For solving a problem, it needs to be precisely defined.
- The definition means, defining the start state, goal state, other valid states and transitions.

Finding the Solution

- After representation of the problem and related knowledge in the suitable format, the appropriate methodology is Chosen which uses the knowledge and transforms the start state to goal state.
- The Techniques of finding the solution are called search techniques.
- Various search techniques are developed for this purpose.



Representation of AI Problem

AI problem can be covered in following four parts:

- Lexical part
- Structural part
- Procedural Part
- Semantic Part



Representation of AI Problem

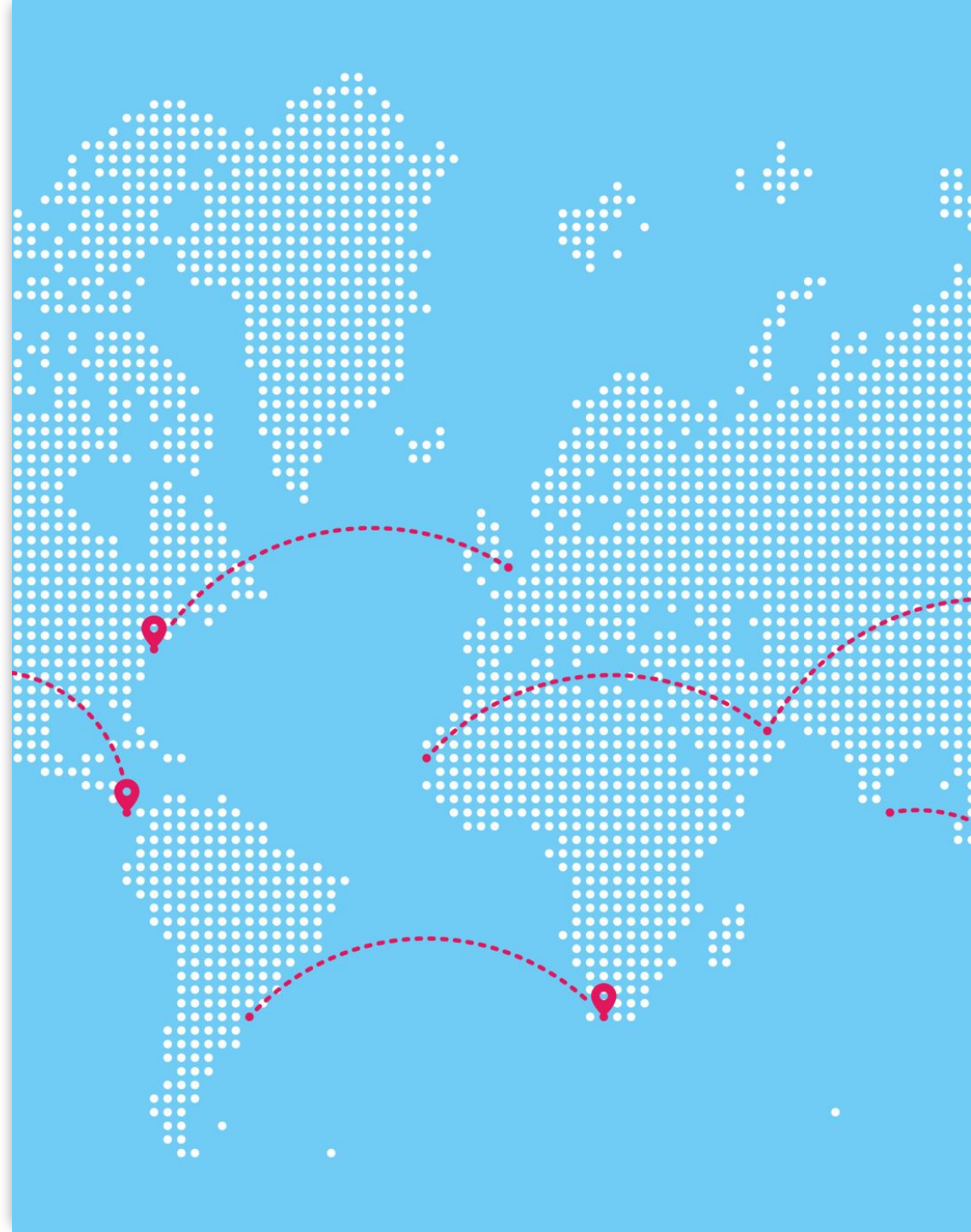
Example: Google Maps Route Finder

- AI Problem → *“Find the shortest route from your current location to your destination.”*

Representation of AI Problem

1. Lexical Part

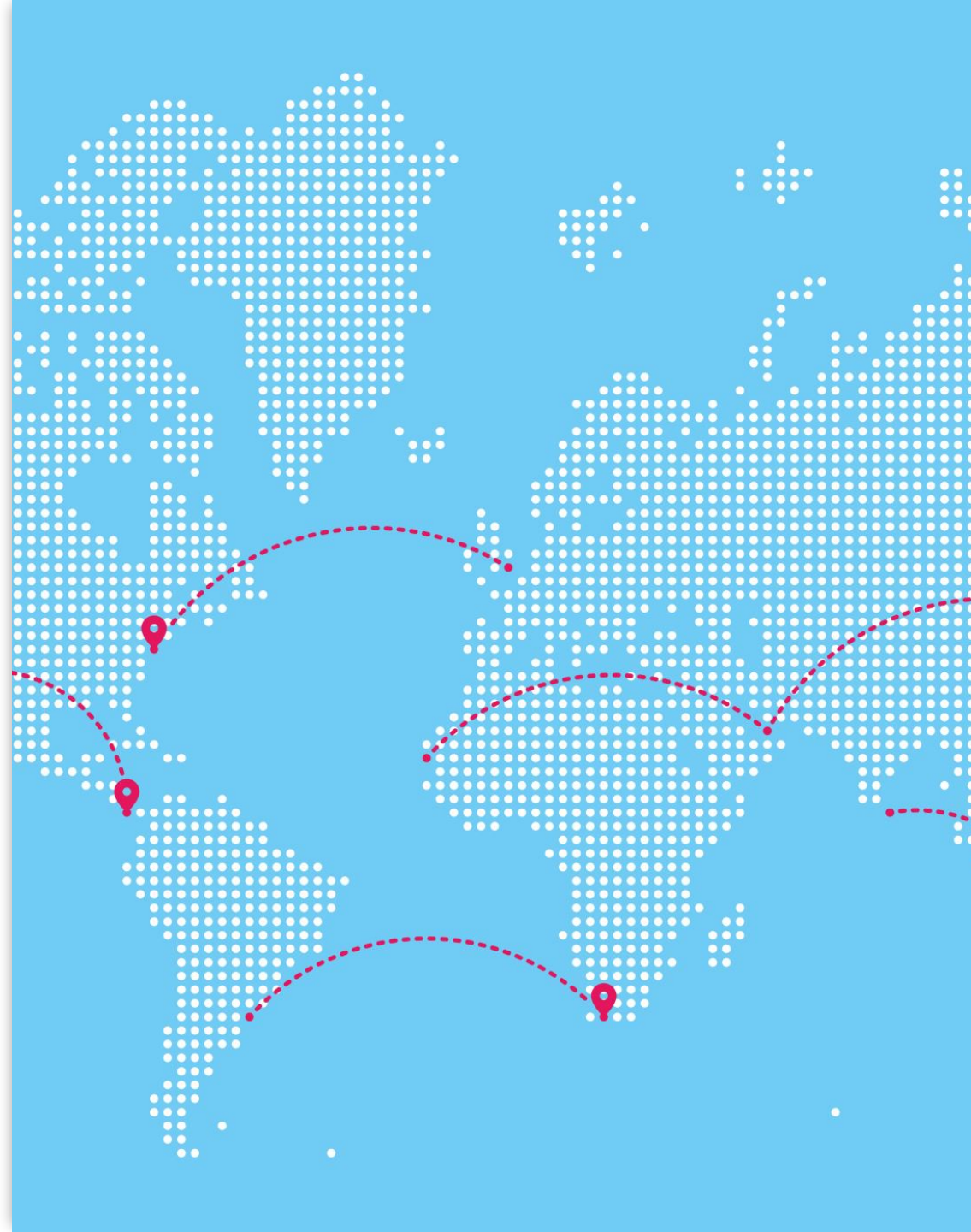
- Determines which **symbols** (basic elements) are allowed in the representation.
- **In this problem:**
The symbols are:
- Names/IDs of locations or intersections (A, B, C, ...)
- Distances or travel times between locations e.g. $\text{distance}(A,B)=5$
- Possible actions like $\text{Move}(A \rightarrow B)$



Representation of AI Problem

2. Structural Part

- Describes **how the symbols are related or arranged** (rules or constraints).
- **In this problem:**
The structure is a **graph**, where:
 - **Nodes** = locations (A, B, C, ...)
 - **Edges** = roads connecting them
 - **Weights** = distances or times
- **Example:**
Road(A,B) and Road(B,C) are valid connections.
But Road(A,Z) is **not valid** if there's no road — that's a structural constraint.



Representation of AI Problem

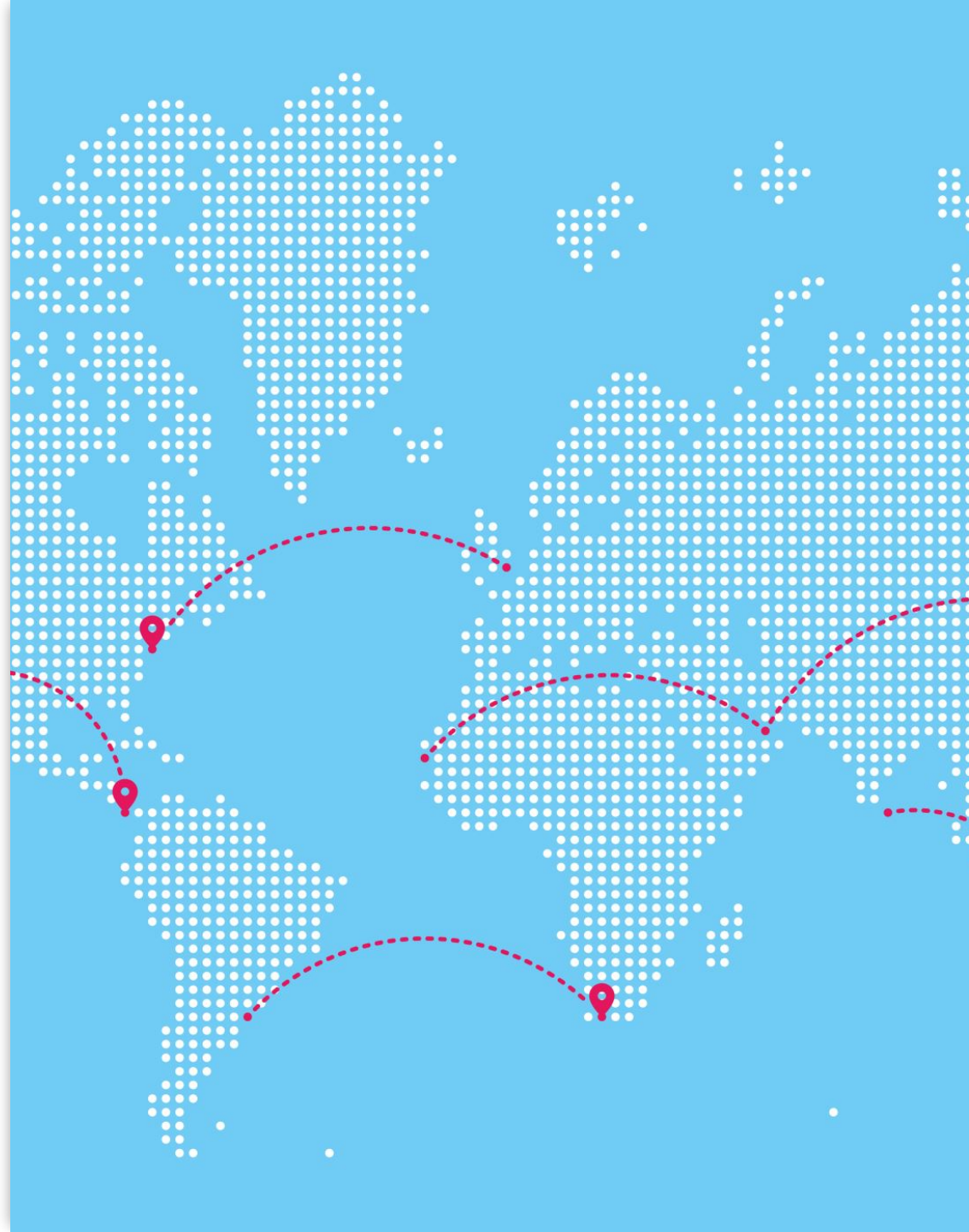
3. Procedural Part

- Describes **how to create, modify, and use** these representations to find a solution.

In this problem:

Procedures include:

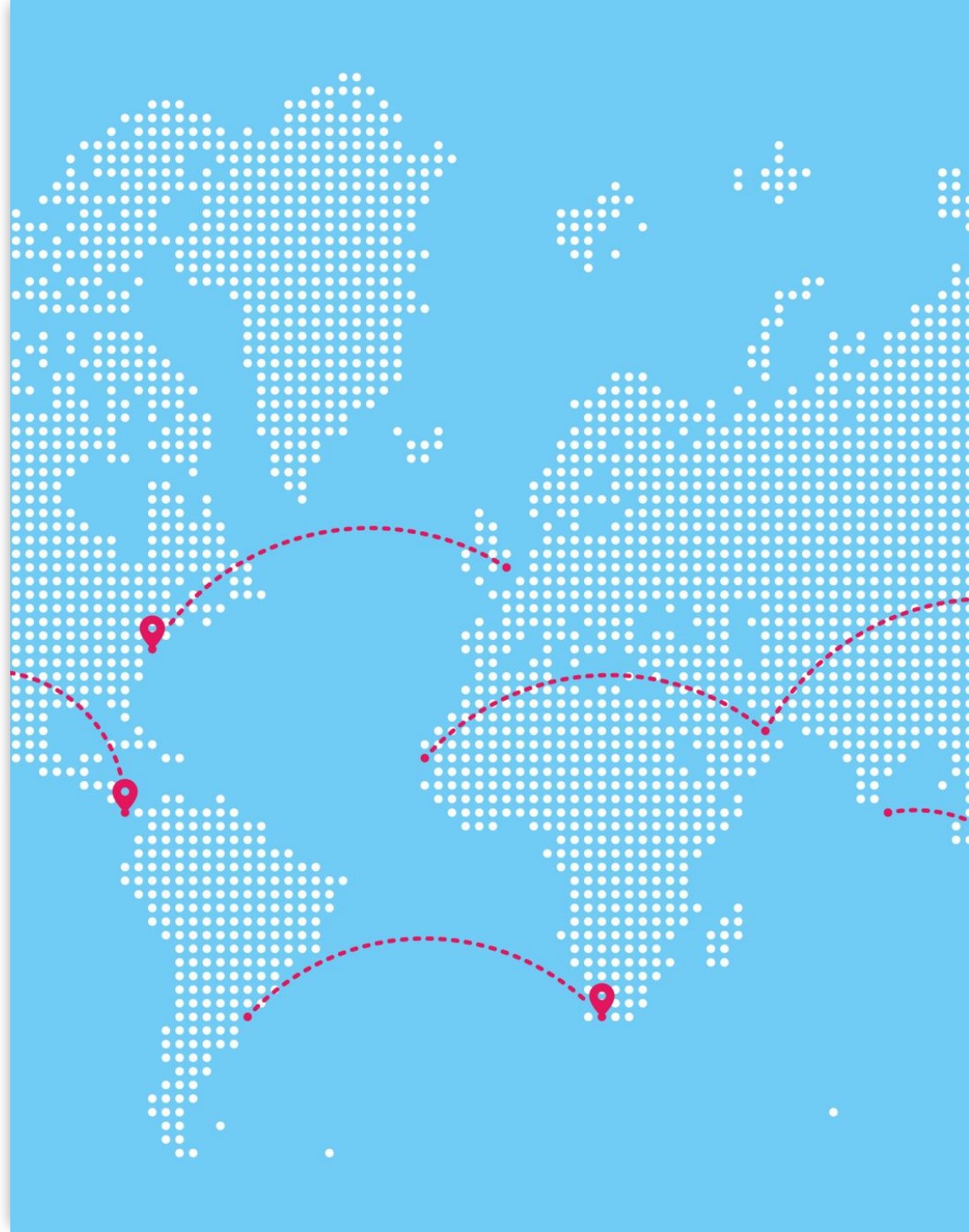
- **Generating** new paths (using algorithms like Dijkstra's or A* search)
- **Updating** routes if traffic changes
- **Selecting** the shortest or fastest route



Representation of AI Problem

4. Semantic Part

- Gives **meaning** to the symbols and structures, what they represent in the real world.
- **In this problem:**
 - The node “A” means a real physical place (e.g., “Main Street”).
 - The edge (A, B) means a real road connecting two points.
 - The weight (5 km) represents an actual distance or travel time.
- **Example:**
If $\text{distance}(A, B) = 5$, the AI understands this means “the physical distance between these two locations is 5 km.”



Representation of AI Problem- Other Examples

AI Problem	Lexical	Structural	Procedural	Semantic
Chess Game	Pieces: King, Queen, etc.	Legal moves, board positions	Minimax algorithm for move selection	Piece meanings (King must be protected)
Speech Recognition	Words, phonemes	Grammar rules	Neural network processing audio	Sound → meaning of spoken language
Medical Diagnosis AI	Symptoms, diseases	Symptom–disease relationships	Inference rules, reasoning engine	“Fever + Cough → Possible Flu”

Components of a Search Problem

**A problem can
be defined
formally by
five
components:**

- **1. Initial State**

“The initial state that the agent starts in.”

Example:

- In a maze → the start cell
- In route planning → your starting city
- In water jug problem → both jugs empty (0,0)

Components of a Search Problem

A problem can
be defined
formally by
five
components:

2. Action (Operators)

“A description of the possible actions available to the agent.”

These are the **moves** or **steps** the agent can take to change the current state into a new one.

Example:

- Maze → move up/down/left/right
- Vacuum → suck dirt or move
- Water jug → fill, empty, pour

Components of a Search Problem

A problem can be defined formally by five components:

- **3. Transition Model**

“Describes how actions change the state.
Represented by $\text{RESULT}(s, a)$.”

The **transition model** tells us **what happens** when an action is applied to a given state i.e., what the *next state* will be.

Example:

- $\text{RESULT}((A, \text{Dirty}, \text{Dirty}), \text{Suck}) \rightarrow (A, \text{Clean}, \text{Dirty})$
- $\text{RESULT}((2, 3), \text{Pour}(A \rightarrow B)) \rightarrow (2 - (3 - 3) = 2, 3)$ (no change if B is already full)

Successor = the new state you get from applying one action.

So, **successor states** = all the possible next states from a given state.

Components of a Search Problem

**A problem can
be defined
formally by
five
components:**

- **4. State Space**

“The set of all possible states that can be reached from the initial state by applying actions.”

It's the **map of all reachable situations** — a **graph** where:

- **Nodes = states**
- **Edges = actions** (transitions between states)
- **Path = sequence of states connected by actions**

Example:

In a maze, the state space is every cell reachable from the start by walking through valid paths.

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Components of a Search Problem

A problem can
be defined
formally by
five
components:

- **5. Goal Test**

“Determines whether a given state is a goal state.”

A **check** to see if the current state satisfies the desired condition (i.e., has the agent solved the problem?).

Example:

- Maze → agent reached the exit cell?
- Water jug → one jug has exactly 2 liters?
- Route planning → reached destination city?
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Components of a Search Problem

A problem can
be defined
formally by
five
components:

- **6. Path Cost Function**

“Assigns a numerical cost to each path. The agent selects a path based on cost.”

Each move or path can have a **cost** (like distance, time, or energy).

The **path cost** is the sum of all action costs.

The agent usually chooses the **lowest-cost path** (optimal solution).

Example:

- In route planning → total distance traveled
- In vacuum → total time or energy used

Example

function SIMPLE-PROBLEM-SOLVING-AGENT(*percept*) **returns** an action

persistent: *seq*, an action sequence, initially empty

state, some description of the current world state

goal, a goal, initially null

problem, a problem formulation

state $\leftarrow$ UPDATE-STATE(*state*, *percept*)

if *seq* is empty **then**

goal $\leftarrow$ FORMULATE-GOAL(*state*)

problem $\leftarrow$ FORMULATE-PROBLEM(*state*, *goal*)

seq $\leftarrow$ SEARCH(*problem*)

if *seq* = failure **then return** a null action

action $\leftarrow$ FIRST(*seq*)

seq $\leftarrow$ REST(*seq*)

return *action*

- A simple problem-solving agent.
- It first formulates a goal and a problem, searches for a sequence of actions that would solve the problem, and then executes the actions one at a time.
- When this is complete, it formulates another goal and starts over.

Formulating problems

- A problem formulation is a model—an abstract way of describing a problem mathematically, not the real-world situation itself.
- An agent's understanding of its actions and current state depends on how it interacts with its environment through its percepts (what it senses) and actions (what it does).
- The more connected the agent is to its environment, the more knowledge it has about its situation.

Formulating problems

Four essentially different types of problems

- **Single-state problems:** The agent knows exactly which state it is in and how its actions affect the state.
- **Multiple-state problems:** The agent is unsure of its exact state and considers multiple possible states at once.
- **Contingency problems:** The agent plans for different possible outcomes, considering unexpected events or changes.
- **Exploration problems:** The agent does not know the environment and must explore to learn about states and actions.



Any Questions